

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EC)

December 2011

EC 302 Integrated circuits & application

Date: 13.12.2011, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) In the given circuit of fig. (1) If the input V_1 & V_2 are to be amplified by the same amplification factor, what should be the value of R? [4]
- (b) Find the op-amp gain of the circuit shown in fig.2. [3]
- Q - 2 (a) Explain op-amp as integrator. Discuss limitation of basic integrator. How it is overcome in practical integrator. [6]
- (b) For the inverting amplifier of fig.3, determine the max. possible O/P offset voltage due to :
- (1) input offset voltage V_{io} [5]
- (2) input bias current I_B
- What value of R_{om} is needed to reduce the effect of i/p bias current I_B ? (consider $V_{io\max}=6\text{ mv}$ & $I_{b\max}=500\text{ na}$)
- (c) Explain the block diagram of typical op-amp. [3]

OR

- Q - 2 (a) Design a peaking amplifier to provide a gain of 5 at a peak frequency of 10kHz. Assume $C=0.05\mu\text{F}$, Coil resistance $R=15\Omega$. [5]
- (b) Write short note on differential amplifier with three op-amps. [5]
- (c) Write short note on antilog Amplifier. [4]
- Q - 3 (a) For triangular wave generator prove that $f_0=R_3/4R_1C_1R_2$. [6]
- (b) Consider non inverting amplifier config. With 3-i/ps as an averaging amplifier, $V_a=V_b=1.5\text{v}$, $V_c=3\text{v}$, $R_1=R=1.5\text{k}\Omega$, $V_0=5.2\text{v}$. Determine the required value of R_f . [4]
- (c) The monostable multivibrator using 555 is to be used as a divide by 4 networks. The frequency of input trigger is 15kHz. if the value of $c=0.01\mu\text{F}$. What should be the value of R_A ? Assume suitable data. [4]

OR

- Q - 3 (a) Design an astable multivibrator having an o/p frequency of 10kHz with a duty cycle of 25%. [5]
- (b) Explain with circuit diagram operation of Schmitt trigger. [5]
- (c) Design and draw circuit diagram to satisfy the following $V_0=(V_1+2V_2+3V_3)/3$ [4]
- Where V_0 =op-amp o/p, V_1, V_2, V_3 =op-amp i/p.

SECTION – II

- Q - 4 (a) Explain performance parameters with reference to voltage regulator. [3]
 (b) Draw the attenuation and magnitude response for the Band pass filter and Band stop filter. [2]
 (c) Explain: (1) Insertion gain (2) insertion Loss. [2]
- Q - 5 (a) Discuss the principle of PLL and explain the Monolithic PLL & Discrete PLL in detail. [8]
 (b) Using LM317, Design an adjustable voltage regulator to satisfy the following specifications: output voltage $V_0=10.5\text{v}$ to 18.75v , output current $I_0=0.5\text{A}$, $I_{ADJ}=100\mu\text{A}$, $R_1=240\Omega$ and $V_{REF}=1.25\text{v}$ [3]
 (c) Using the 7805C voltage regulator, design a current source that will deliver 150mA current to the 8Ω , load. Draw the complete schematic diagram of the circuit. [3]

OR

- Q - 5 (a) Specifications for Butterworth low pass filter are: $\alpha_{\max}=0.50\text{dB}$ $\alpha_{\min}=20\text{dB}$, $\omega_p=1000$ rad/sec, $\omega_s=2000\text{rad/sec}$. Determine n , half power frequency ω_0 , and actual attenuation at the edge of pass band $\alpha(\omega_p)$ and stop band $\alpha(\omega_s)$. [6]
 (b) Draw Sallen Key low pass filter circuit and determine its transfer function. Obtain the sallen key high pass filter circuit by applying RC-CR transformation on the sallen key low pass filter circuit with $K=1$ and $R_1=R_2=1$. [8]
- Q - 6 (a) From the generalized transfer function of low pass filter implement the Biquad circuit with necessary equations. [7]
 (b) Design a circuit to realize the transfer function of the following Bode plot given in fig.4. Use inverting configuration. [7]

OR

- Q - 6 (a) Explain Active All Pass filter with necessary equations. Draw its magnitude & phase response. [7]
 (b) Derive element scaling equation for L_{new} , C_{new} & K_{new} in terms of K_m , K_f & the old elements. [7]

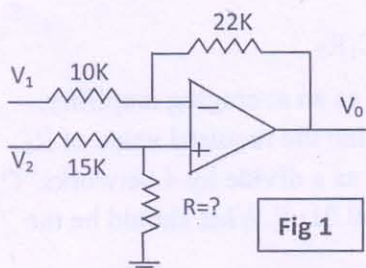


Fig 1

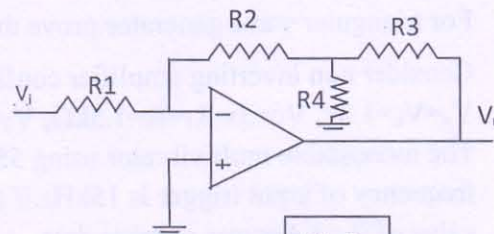


Fig 2

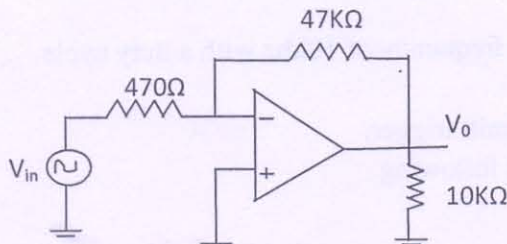


Fig 3

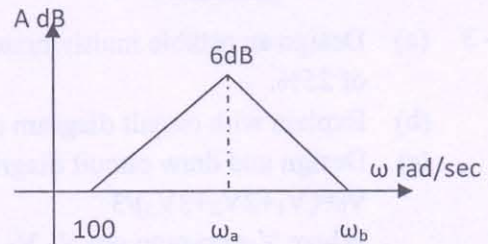


Fig 4